

CFB BORDEN, ON, Canada

WMO: 715340

Lat: 44.270N

Lon: 79.910W

Elev: 223

StdP: 98.68

Time Zone: -5.00 (NAE)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.3	-20.4	-28.7	0.3	-23.6	-24.3	0.4	-19.7	11.1	-3.8	10.0	-2.7	1.3	340	0.583

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.3	31.7	22.2	29.8	21.2	28.2	20.3	23.4	29.3	22.5	27.9	21.5	26.4	5.4	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.6	16.7	26.0	20.7	15.8	25.2	19.8	14.9	24.3	70.8	29.2	67.2	28.0	63.8	26.6	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
9.6	8.4	7.5	-32.1	34.3	3.1	1.5	-34.3	35.4	-36.1	36.3	-37.9	37.1	-40.1	38.3	
			DB												
			WB	-32.0	25.1	3.4	0.9	-34.4	25.8	-36.4	26.3	-38.3	26.8	-40.7	27.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.8	-7.3	-6.9	-1.8	5.0	12.2	17.3	20.0	19.2	15.4	8.6	2.5	-3.5
	DBStd	11.00	7.17	6.37	6.31	5.20	5.06	4.07	3.47	3.28	4.47	4.74	5.21	5.70
	HDD10.0	2314	535	474	367	168	32	1	0	0	6	84	229	418
	HDD18.3	4411	793	707	623	402	202	68	22	29	111	303	474	677
	CDD10.0	1147	0	0	2	17	101	219	311	284	167	42	4	0
	CDD18.3	203	0	0	0	1	12	36	74	55	23	2	0	0
	CDH23.3	2373	0	0	2	19	179	460	829	591	269	24	0	0
	CDH26.7	730	0	0	0	3	53	154	268	169	79	3	0	0
Wind	WSAvg	3.2	3.6	3.6	3.5	3.7	3.2	2.9	2.7	2.6	2.7	3.2	3.3	3.5
Precipitation	PrecAvg	834	60	52	56	66	73	78	81	77	76	72	73	69
	PrecMax	1062	104	106	144	131	132	207	119	168	135	134	136	108
	PrecMin	623	28	26	18	22	26	21	33	35	30	23	25	30
	PrecStd	98	20	23	27	27	25	41	27	29	25	28	27	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.9	10.4	19.5	26.3	30.7	33.1	33.9	33.3	32.0	26.7	18.0	12.0
		MCWB	8.2	7.6	14.5	16.1	20.2	22.6	23.7	22.5	22.3	18.9	12.6	10.6
	2%	DB	6.1	6.5	13.9	21.3	28.1	30.6	31.6	30.4	29.0	22.8	15.0	8.2
		MCWB	4.5	4.8	9.9	13.8	19.1	21.3	22.1	21.7	21.0	17.1	11.8	6.7
	5%	DB	4.0	3.7	10.4	17.9	25.0	28.2	29.5	28.5	26.6	19.4	12.9	5.5
		MCWB	2.7	1.8	6.8	11.5	17.3	20.2	21.1	20.8	19.7	15.3	10.1	4.1
	10%	DB	2.1	1.8	7.1	14.6	22.3	25.9	27.7	26.7	24.0	16.7	10.4	3.6
		MCWB	1.1	0.3	4.3	9.3	15.3	18.9	20.1	19.8	18.3	13.5	8.2	2.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.7	8.2	15.3	17.7	21.8	23.9	24.8	24.7	23.6	20.8	14.2	10.8
		MCDB	9.3	9.4	18.9	23.6	28.0	30.4	31.3	30.1	29.0	24.6	16.0	12.1
	2%	WB	5.0	5.0	10.5	14.9	20.2	22.6	23.4	22.9	22.2	17.8	12.4	6.8
		MCDB	5.8	6.7	12.9	20.4	26.6	29.2	29.4	28.4	27.0	21.2	14.7	7.8
	5%	WB	2.7	2.0	7.5	12.4	18.5	21.2	22.4	21.9	20.7	15.9	10.3	4.4
		MCDB	3.7	3.2	10.1	16.5	23.5	26.5	27.6	26.8	24.8	18.7	12.4	5.4
	10%	WB	1.0	0.4	4.5	9.8	16.5	19.8	21.4	20.8	19.3	14.0	8.4	2.4
		MCDB	2.0	1.7	6.7	14.2	21.0	24.4	26.2	24.9	22.9	16.4	10.5	3.4
Mean Daily Temperature Range	5% DB	MDBR	9.2	9.8	10.5	12.2	14.1	13.3	13.3	13.2	13.4	10.4	8.9	7.5
		MCDBR	8.3	10.6	14.9	18.4	17.4	16.2	15.4	15.6	16.1	15.2	12.5	8.7
		MCWBR	7.5	9.4	10.9	11.6	9.5	8.4	7.5	7.7	8.6	9.7	9.6	7.6
	5% WB	MCDBR	8.0	9.8	13.8	15.6	15.3	13.9	13.1	12.7	13.3	13.0	11.3	8.6
		MCWBR	7.6	9.0	10.6	11.0	8.8	7.8	6.8	6.9	7.7	8.8	9.3	7.9
Clear-Sky Solar Irradiance	taub	0.283	0.301	0.318	0.357	0.387	0.410	0.404	0.396	0.364	0.345	0.310	0.283	
	taud	2.363	2.292	2.401	2.349	2.317	2.275	2.311	2.319	2.408	2.446	2.490	2.425	
	Ebn at Noon	842	886	921	906	883	861	861	856	859	828	808	808	
	Edh at Noon	77	99	102	117	125	131	125	120	101	86	69	66	
All-Sky Solar Radiation	RadAvg	1.35	2.23	3.52	4.48	5.39	5.89	5.95	5.12	4.04	2.37	1.41	0.98	
	RadStd	0.15	0.28	0.37	0.45	0.54	0.43	0.38	0.27	0.38	0.25	0.20	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page